

Neurodevelopmental and Behavioral Problems in Children and School Dysfunction

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Session Objectives

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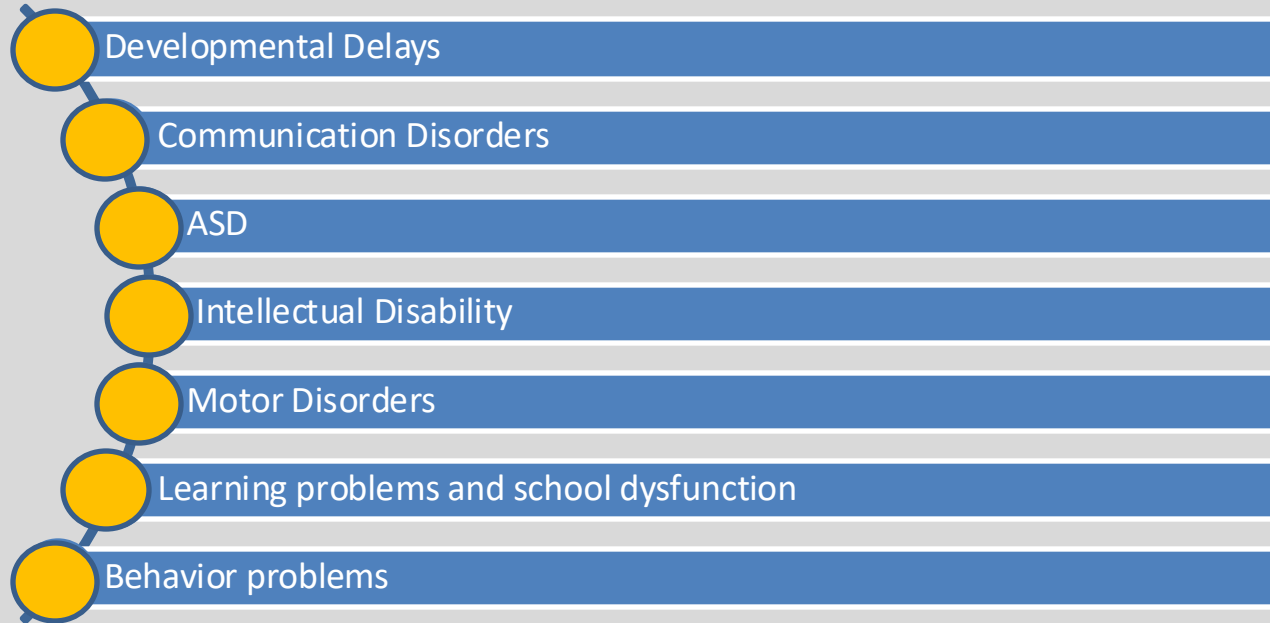
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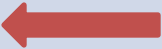
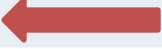
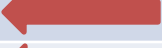
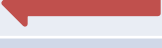
- Compare and contrast the diagnostic features for speech delay, the communication disorders, and autism spectrum disorder, providing a differential diagnosis, workup and management for each.
- Define intellectual disability, identify potential causes, and describe the workup for a child suspected of having intellectual disability.
- Compare and contrast chronic tic disorder and Tourette syndrome.
- Explain the definition of learning disorder, describe the important elements of the history and workup for a child presenting with a suspected learning disorder and outline the differential diagnosis.
- Compare and contrast ODD and CD and outline the elements of the history that are important for a child presenting with a behavior problem.

OUTLINE

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Condition	Prevalence
Slower learner (IQ ≤89)	23% 
Asthma	8.4%
Learning disability	7.7% 
ADHD	6.7% 
Other developmental delay	4.4% 
Autism spectrum disorder	1.7% 
Epilepsy	1%
Congenital heart disease	1%
Cerebral palsy	0.4%
Inflammatory bowel disease	0.4%
Juvenile rheumatoid arthritis	0.4%
Diabetes	0.2%
Cancer	0.02%
Cystic fibrosis	0.04%
Chronic renal disease	0.008%

Prevalence of Developmental- Behavioral Disorders and Other Chronic Medical Conditions in Children

Distribution of Disability Types among students 3-21 years of age served under the Individuals with Disabilities Education Act (IDEA)

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Disability type	Percent
Specific learning disability ¹	32
Speech or language impairment ²	19
Other health impairment ³	15
Autism	13
Developmental delay ⁴	7
Intellectual disability	6
Emotional disturbance	4
Multiple disabilities	2
Hearing impairment	1

Patient 1

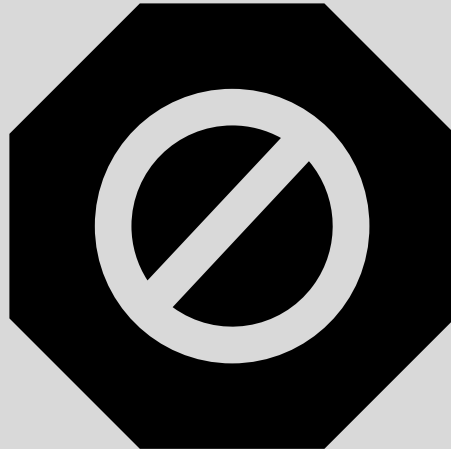
- 24 month old male at your office for a well visit
- Mother's concerns:
 - Child not saying as many words as sibling
 - Child has frequent tantrums
- Birth Hx: prematurity at 34 weeks of gestation
- FH: learning problems, ADHD
- Exam: normal growth parameters and physical exam
- Observations during encounter: poor eye contact at times, does not respond to name, only says "no", plays with toy car

Differential Diagnosis

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Pause video here and write out a differential diagnosis for this child. Include at least three diagnoses and put the most likely diagnosis first.



Patient 1: Differential Diagnosis

- Expressive speech delay
- Communication disorder
- Hearing loss
- Autism spectrum disorder
- Intellectual disability

Often, this group
of disorders will be
on your
differential
diagnosis list
together

What is a Developmental Delay?

Developmental delay: failure to meet milestones

- Isolated delay
 - Motor
 - Speech and language
 - Cognitive
 - Social
- Global delay: delay in $\geq$ two domains
- Delay vs. disorder – related to function and degree of impairment

Red Flags

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Table 2. **Motor Red Flags**

Age	Red Flag
4 months	Lack of steady head control while sitting
9 months	Inability to sit
18 months	Inability to walk independently

Table 2. **Cognitive Red Flags**

Age	Red Flag
2 months	Lack of fixation
4 months	Lack of visual tracking
6 months	Failure to turn to sound or voice
9 months	Lack of babbling consonant sounds
24 months	Failure to use single words
36 months	Failure to speak in three-word sentences

Table. **Social-Emotional Red Flags**

Age	Red Flag
6 mo	Lack of smiles or other joyful expressions
9 mo	Lack of reciprocal (back-and-forth sharing of) vocalizations, smiles, or other facial expressions
12 mo	Failure to respond to name when called Absence of babbling
	Lack of reciprocal gestures (showing, reaching, waving)
15 mo	Lack of proto-declarative pointing or other showing gestures
	Lack of single words
18 mo	Lack of simple pretend play
	Lack of spoken language/gesture combinations
24 mo	Lack of two-word meaningful phrases (without imitating or repeating)
Any age	Loss of previously acquired babbling, speech, or social skills

 Loss of milestones is always a red flag

Management of Developmental Delay

- **Intervene early**
- Refer for appropriate services
 - Speech therapy
 - Physical therapy
 - Occupational therapy
- Refer for subspecialty evaluation
- Support and education for parents
- Provide a medical home

Speech and Language Development

- Language development: early marker of cognitive development
- Delays often associated with comorbidity
 - Learning problems
 - Behavior problems
- Exposure to speech matters

Communication Disorders – Diagnostic Features

Language
Disorder

Speech Sound
Disorder

Child-Onset
Fluency Disorder

Social (Pragmatic)
Communication
Disorder

Language disorder – persistent difficulties in comprehension and/or production of language

- Results in functional limitations
- Presents as reduced vocabulary, limited sentence structure and limited conversation
- Deficits in spoken and/or written communication or sign language.
- Can involve expressive and/or receptive abilities.

Communication Disorders – Diagnostic Features

Language
Disorder

Speech Sound
Disorder

Child-Onset
Fluency Disorder

Social (Pragmatic)
Communication
Disorder

Speech sound disorder: persistent difficulty with speech and sound production that interferes with speech intelligibility or prevents verbal communication

- Articulation of speech sound is impaired to the point that it interferes with ability to communicate

*Speech sound should be 100% intelligible by 4 years of age

Communication Disorders – Diagnostic Features

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Language
Disorder

Speech Sound
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Child-Onset
Fluency Disorder

Social (Pragmatic)
Communication
Disorder

Child-onset fluency disorder: stuttering

- Clinically significant impairment in speech fluency and timing

Communication Disorders – Diagnostic Features

Language
Disorder

Speech Sound
Disorder

Child-Onset
Fluency Disorder

Social (Pragmatic)
Communication
Disorder

Social pragmatic communication disorder: impairment in the use of language for social exchanges, ability to adapt communication style to context of interaction, ability to follow conventional and cultural norms for conversation, ability to understand ambiguous language.

- Differentiate from autism

Speech & Language Disorders: Work-up and Management

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Autism Spectrum Disorder

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A lifelong neurodevelopmental disorder associated with deficits in social communication and social interaction and accompanied by restrictive/repetitive behaviors.

Risks for Autism Spectrum Disorder

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Genetics

- Chromosomal disorders
- Heredity

Biological

- Older parents
- Prenatal exposures
- Prematurity
- Low birth weight

Vaccines do NOT cause autism

Autism Spectrum Disorder: Two Core Dimensions

SOCIAL

1. Deficits in social communication and social interaction

NON-SOCIAL

2. Restrictive, repetitive behaviors, interests or activities

ASD:

Diagnostic Features

Deficits in social communication and social interaction

Social-emotional reciprocity

Nonverbal communication behaviors

Developing, maintaining, understanding relationships

Need at least 1 from each social category

Restrictive, repetitive behaviors, interests or activities

Stereotyped, repetitive movements or speech

Insistence on sameness, inflexible adherence to routines, ritualized behavior

Restricted interests

Hyper or hypo-reactivity to sensory input

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Cognitive development:
most individuals with ASD
have intellectual abilities in
the normal range (IQ > 70)

Need at least 1 from 2 of 4 non-social categories

Early Symptoms of ASD

- Lack of social interaction
 - Response to name - [Lack of Response to Name](#)
 - Eye contact - [Lack of Eye Contact](#)
 - Social smile
 - Limited joint attention - [Lack of joint attention](#)
 - “Failure to connect” with parents- [Comparison of Twins](#)
- Delay in language development
 - Social babbling
 - Verbal comprehension
 - Language regression
- Lack of gestures to communicate (pointing)
- [Echolalia](#)
- Limited imaginative play; odd play patterns - [Inappropriate Play](#)
- Atypical motor mannerisms

Diagnosing ASD

- Early diagnosis and intervention is optimal
 - History
 - Assess hearing and vision
 - Consider comorbid conditions
- Screening tools used by PCP
 - Infant-Child Checklist: < 18 months
 - **M-CHAT-R**: 18 – 30 months
 - Child Autism Screening test: preschool/school age
- Refer for in depth evaluation by Developmental Pediatrician and multidisciplinary team
 - **Do not delay management steps while awaiting this evaluation and formal diagnosis**
 - Refer to appropriate services **ASAP**

Management of ASD

- Educational and behavioral interventions
 - Applied Behavioral Analysis (ABA)
 - Speech therapies
 - Functional communication
 - Occupational therapies
 - Social skill support
- Medications for specific symptoms – inattention, anxiety, etc.
- Ongoing monitoring for comorbidities
- Support for family

Speech Disorder Vs. Autism

TABLE 4. **Comparison of Specific Language Impairment (SLI), Social (Pragmatic) Communication Disorder (SPCD), and Autism Spectrum Disorder (ASD)**

	SLI	SPCD	ASD
Social	• Strong interest • Joint attention • Affective reciprocity • Difficulties may be due to deficits in basic language abilities	• Strong interest • Inappropriate interactions that are not due to deficits in basic language abilities	• Little interest • Limited/absent joint attention • Limited/absent affective reciprocity
Play	• Imitation • Pretending	• Imitation • Pretending	• Very limited imitation • No symbolic play • Attachment to odd objects
Routines	• No restricted, repetitive patterns of behavior, interests, or activities	• No restricted, repetitive patterns of behavior, interests, or activities	• Rigid and ritualized behavior that does not seem to serve any functional purpose
Behavior Patterns	• Frustrated because of lack of understanding or inability to communicate	• Frustrated because of lack of success in social interactions	• Stereotypic behaviors • Unusual responses to stimuli

Intellectual Disability

Combination of impairment of cognitive **and** adaptive development

Cognitive: Reduced IQ

- Mild
- Moderate
- Severe
- Profound

Adaptive: Deficits in function

- Conceptual
- Social
- Practical skills

Causes of ID

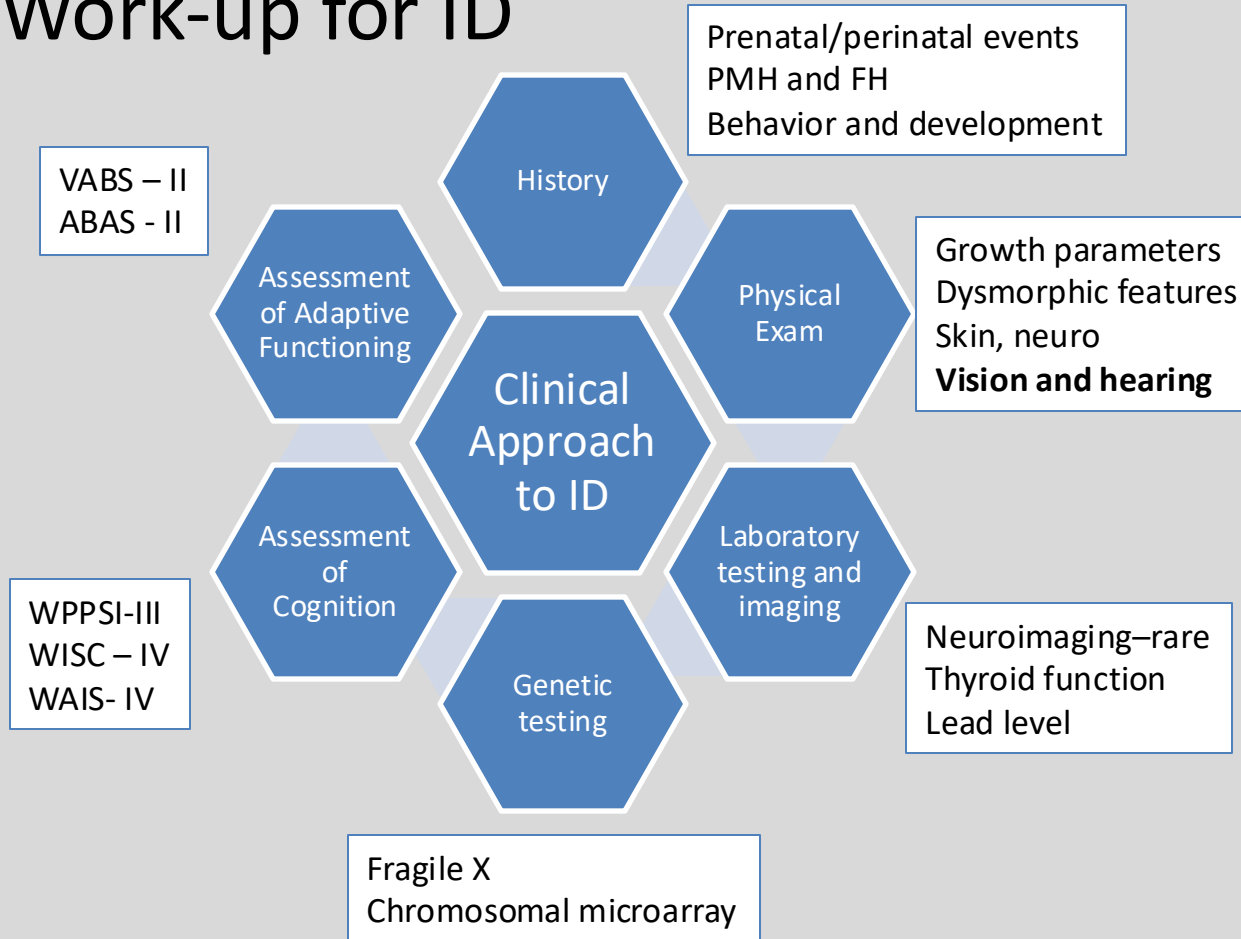
Table 1. Causes of Intellectual Disability

Prenatal	Infection (eg, toxoplasmosis, rubella, cytomegalovirus, herpes simplex virus)
	Genetic abnormalities
	Trisomy 21
	Fragile X
	Other (point mutations; chromosome deletions, duplications, rearrangements, etc)
	Toxins and teratogens
	Alcohol
	Drugs
	Radiation
	Congenital hypothyroidism
	Inborn errors of metabolism
Perinatal	Hypoxia
	Complications of prematurity
	Intracranial hemorrhage
	Perinatal CNS infection
Postnatal	Acquired brain injury
	CNS hemorrhage
	CNS infection
	CNS malignancy
	Severe environmental deprivation
	Severe malnutrition
	Toxins (lead, mercury)

CNS, central nervous system.

- Prenatal and genetic most common
 - Fragile X
 - Trisomy 21
 - Fetal alcohol spectrum disorder
 - Infection
- Often no cause identified

Work-up for ID



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Management/Supports for ID

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Educational Supports

- IDEA
- Educational inclusion
- Special education
- IEP

Skill Building

- Communication
- Self-help
- Social
- Vocational

Monitor for Comorbidities

- ADHD
- Autism
- Sensory impairment
- Cerebral palsy
- Psychiatric/behavioral disturbance
- Sleep disorders

Motor Disorders

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Developmental Coordination Disorder

- Coordinated motor skills
- Clumsiness, slowness, inaccuracy

Stereotypical Movement Disorder

- Repetitive, purposeless motor behaviors
- May or may not be self-injurious

Tourette Syndrome

- Multiple motor AND one or more vocal tics
- Present for at least one year

Persistent/Chronic Tic Disorder

- Motor OR vocal tics
- Present for at least one year

Return to Patient 1

- Developmental screening with Ages & Stages demonstrated a delay in communication
- M-CHAT-R was not concerning for ASD
- Referred for audiology testing and formal speech and language evaluation
 - Demonstrated bilateral hearing loss
 - Speech and language evaluation confirmed delayed speech
 - Sent to ENT for further evaluation and hearing aids
 - Speech therapy started
 - At his 30 month well child visit he demonstrated improvement

Patient 2

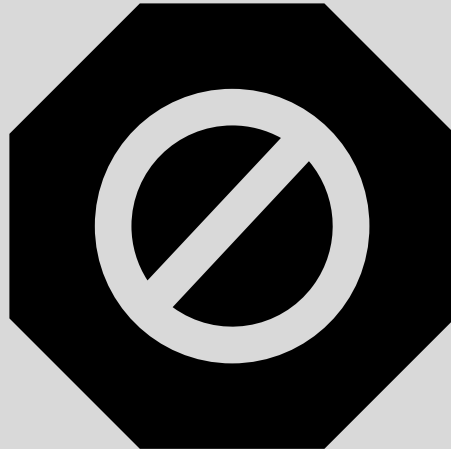
- 7 year old male in second grade brought to you specifically because of school problems
 - Poor grades – C, D, F
 - Teacher recommends he repeat
 - For last several months: getting in trouble at school for talking back, getting out of seat, not listening, fighting with peers
 - Birth Hx: Prematurity
 - PMH: required speech therapy for speech delay
 - Fam Hx: ADHD, learning problems
 - Social Hx: several stressors

Differential Diagnosis

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Pause video here and write out a differential diagnosis for this child. Include at least three diagnoses and put the most likely diagnosis first.



Differential Diagnosis for Patient 2

- Learning disorder
- Behavior problem
 - ODD
 - Conduct disorder
- ADHD
- Intellectual disability
- Mood disorder
- Reaction to psychosocial stressors

Differential
diagnosis for
school
dysfunction

Learning Disorder

Discrepancy between intellectual ability
and academic performance

NOT due to:

- Vision or hearing problems
- Motor disability
- Environmental, cultural or economic disadvantage
- Intellectual disability

Risk factors:

- Prematurity
- Positive family history
- Preschool developmental problems
 - Especially speech and language: *high association with reading LD

Learning Disorder

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Difficulty learning and using academic skills

Reading

- Word reading
- Comprehension

Writing

- Spelling
- Written expression

Mathematics

- Number sense
- Calculations
- Mathematical reasoning

Table 1. Signs and Symptoms of Learning Disability and School Failure

Increased Learning Effort

"School is boring"
School anxiety
Class clown behavior
Spending a much longer time
completing homework
than classmates

School Distress

Frequent failing grades
Frequent absences
Social disengagement
Frequent detention
Suspensions
Aggression and
bullying behaviors

School Failure

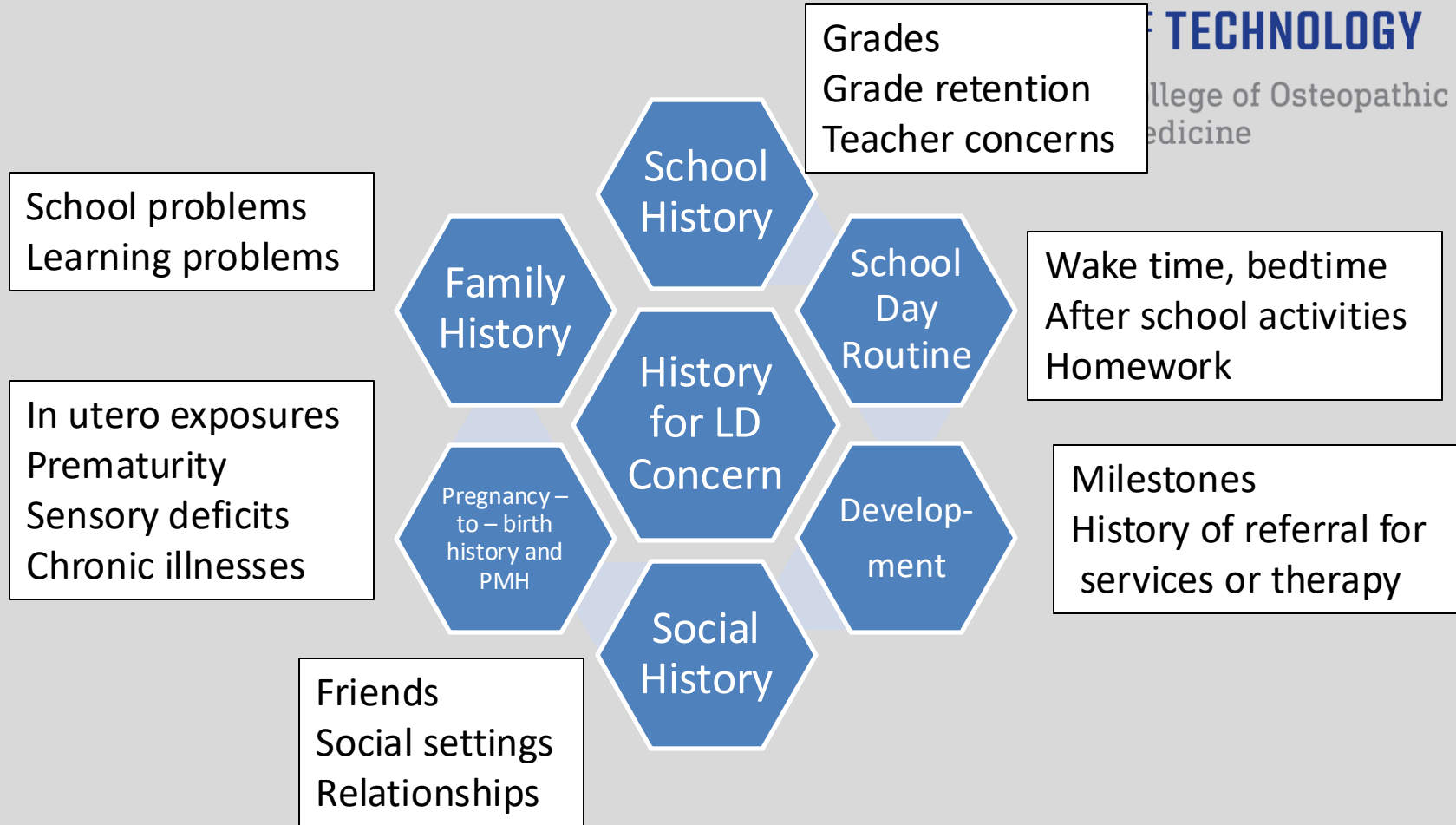
Retention
Expulsion
Dropping out

- Often detected late
- Children try to compensate
- Goal is to detect and intervene early in order to prevent school failure

History is Key to Diagnosis

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Work-Up is Multidisciplinary

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Physical Exam

Dysmorphia

Skin lesions

Neurological
exam

Diagnostic testing

Senses: vision
and hearing

Neuroimaging

Genetic tests

Psycho- educational Testing

Cognitive
assessment

Achievement
testing

Ongoing Management Can Be Complex

- Advocate for family
 - Ensure proper services received
- Provide a medical home
 - Ongoing assessment for functional or mental health problems
 - Review IEP (Individualized Education Program), school accommodations, therapies, report cards
 - Ongoing behavioral and developmental surveillance

Behavior Problems

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	Oppositional Defiant Disorder (ODD)	Conduct Disorder (CD)
Onset	Usually before age 8 y, no later than early adolescence	Child onset before age 10 y Adolescent onset after age 10 y but rarely after age 16 y
Symptoms	Trouble with authority Breaks school rules Always seems angry Argumentative Easily annoyed Noncompliant	Significant aggression toward others Animal cruelty Destruction of property Theft Setting fires Serious violation of rules Law breaking Substance abuse Poor school performance
Duration of Symptoms for Diagnosis	For at least 6 mo	3+ incidents within the last 12 mo with at least one in the past 6 mo

Co-morbid Disorders are Common with ODD and CD

- ADHD
- Learning disorder
- Anxiety
- Depression
- Decreased IQ
- Language deficit
- Impaired executive functioning
- Reaction to psychosocial stressor

Important History for Behavior Problems

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Characteristics
of the
behavior

Child's
development

Temperament
of child and
parents

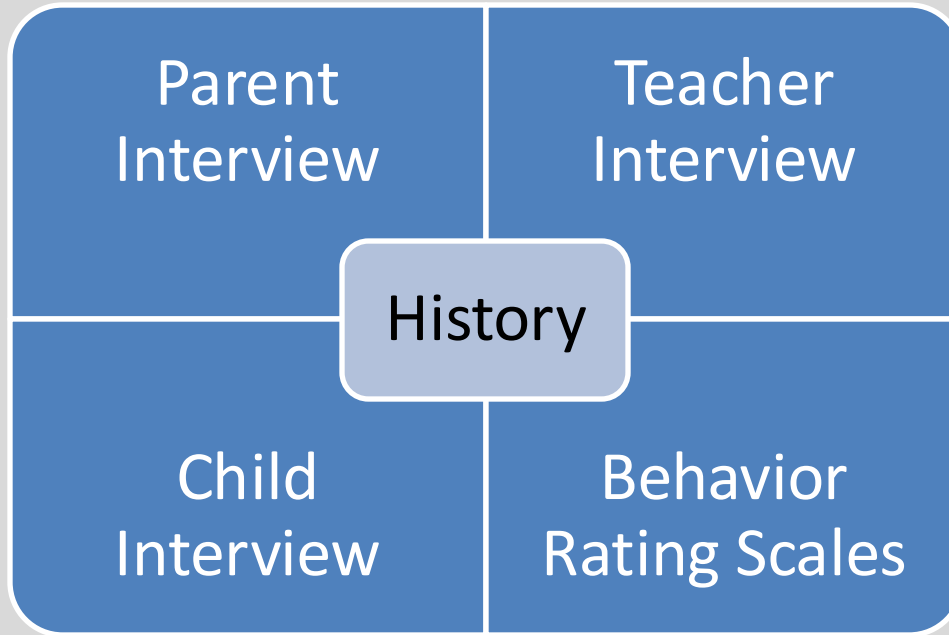
Psychosocial
stressors

Impact on
functioning
and academics

Work-up for Behavior Problems

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- Youth Self Report
- Child Behavior Checklist
- Conner's Rating Scale

Treatment of Behavior Problems

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Parent
Behavior
Management

Cognitive
Behavior
Management

Medication

Patient 2

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 - Teacher recommends he repeat
 - Getting in trouble at school for talking back, getting out of seat, not listening, fighting with peers
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- Compare and contrast ODD and CD and outline the elements of the history that are important for a child presenting with a behavior problem

Summary Slide

- Normal child development is variable
- Delays can occur in any developmental domain
- It is important to monitor closely for developmental delay and neurodevelopmental disorders because intervening early is key
- Always consider a broad differential diagnosis and comorbid conditions
- A thorough history is key
- Education and support for parents is very important

References

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- Boyle CA et al. *Pediatrics* 2011;127:1034-1042
- Balo J et al. *MMWR Surveil Summ* 2018; 67: 1-23
- Shea, Sarah. “Intellectual Disability (Mental Retardation.” *Pediatrics in Review* 33 (2012):110 – 121.
- McQuiston, Susan and Nancy Kloczko. “Speech and Language Development: Monitoring Process and Problems” *Pediatrics in Review* 32 (2011):230 – 239.
- Simms, Mark and Xing Ming Jin. “Autism, Language Disorder, and Social (Pragmatic) Communication Disorder: DSM-V and Differential Diagnosis.” *Pediatrics in Review* 38 (2015):355 – 363.
- Rimrodt, Sheryl, and Paul Lipkin. “Learning Disabilities and School Failure.” *Pediatrics in Review* 32 (2011):315 – 324.
- Zahrt, Dawn and Marlene Melzer-Lange. “Aggressive Behavior in Children and Adolescents.” *Pediatrics in Review* 32 (2011):325 – 332.
- American Psychiatric Association. “Neurodevelopmental Disorders.” in *Diagnostic and Statistical Manual of Mental Disorders, Fifth edition*
- Maenner MJ, Warren Z, Williams AR, et al. Prevalence and Characteristics of Autism Spectrum Disorder Among Children Aged 8 Years — Autism and Developmental Disabilities Monitoring Network, 11 Sites, United States, 2020. *MMWR Surveil Summ* 2023;72(No. SS-2):1–14.
DOI: <http://dx.doi.org/10.15585/mmwr.ss7202a1>.
- National Center for Education Statistics. (2024). Students With Disabilities. *Condition of Education*. U.S. Department of Education, Institute of Education Sciences. Retrieved [date],
from <https://nces.ed.gov/programs/coe/indicator/cgg>.

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